Evidentiary reasoning within the Analytical Hierarchy Process: Ranking decision elements by graph entropy

unpublished working paper

GitHub: hamlett-neil-ur/ahp-graph-entropy

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Updated: July 10, 2026

Abstract

Decision makers often require insight into the relative strength of the elements of evidence on which their choices hinge. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of elements' relative strength. Comparative judgements ("pairwise comparisons") between elements of a given hierarchy model produce lists of edges for a complete graph. Under certain conditions, such graphs are also directed and acyclic, belonging to the class of tournament graphs. Intensities of Importance from comparative judgements provide weights for graphs' edges. Relative-entropy calculations based on Laplacian matrices for induced subgraphs quantify the relative evidentiary strength associated with the decisional evidence ommitted from them. Motivated by quantum-computing research, recent innovations in matrix representations of graphs support the idiosyncratic structures arising from AHP comparative judgements. A previously published case study related to an offshore-pipeline material selection demonstrates the approach. The graph-entropy method provides greater strength-of-contribution contrast than sensitivity analysis from the case study.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

> "...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to *a pretence of exact knowledge that is likely to be false...*" (emphasis added, von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge confronting multiple facets. We conventionally decompose uncertainty into *epistemic* and *aleatory* (or *stochastic*) components (Salicone & Prioli (2018)). Sound evidentiary reasoning must account for the consequences of both.

The present study focuses primarily on the epistemic. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Townsend, et al (2025); Ehrig, et al (2026), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is our evidence?

Ehrig, et al (2026) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of *very small samples of data*" (emphasis added). Circumstances demanding this sort of reasoning arise in numerous contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, all entrepreneurial ventures amount to *epistemic* explorations (Zellweger & Zenger (2023); Camuffo, et al (2024a)). Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when trying to select optimum scenarios.

Nontrivial challenges inevitably arise *even* when considerable amounts of statistically coherent, phenomenologically illuminating empirical data exist. Cover & Thomas (2006) derive an information-theoretic *data-processing inequality*: "...no processing of Y ... can increase the information that Y contains about X." Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive *nonstationarity* in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2025) articulate the case for "...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems..." for many activities.

Extending Dempster's (1967) seminal concept, Shafer (1976) offered a rigorous framework to characterize epistemic uncertainty in evidentiary reasoning. More-recently, Shenoy (2023) applied Shafer's reasoning approach to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024b); Ehrig, et al (2026)).

Hué, et al (2026) applied a stochastic-detection area-under-the-curve (AuC) statistic (*cf.* Fawcett (2026)) to ascertain evidentiary elements' relative influence. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) borrows from sheaf theory (Rosiak (2022)) to characterize global epistemic alignment.

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for "...logic and reasoning with the aid of very small samples of data" (Ehrig, et al (2026)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible *evidentiary span*: The range of measurable factors — *elements* in the vernacular of Saaty (1986, 1994) — believed to differentiate within a discrete, finite range of scenarios under consideration. A *hierarchy model* represents such spans as tree graphs. The responsible individual decision maker or a panel of experts then performs *comparative judgements*: "...the numerical representation of a relationship between two elements that share a common parent" in a tree graph (Saaty (1994)). A subsequent section presents a case study demonstrating the mechanics of AHP to rank plausible alternatives.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025) and Sáenz-Royo & Chiclana (2026) studied methods to improve the rubustness of expert inputs into comparative judgements. Syed, Shabarchin, Lawryshyn (2020) describe AHP's use in eliciting prior probabilities for Bayesian reliability analysis.

Evidentiary-strength analysis using the AHP.

That comparative judgements for an AHP hierarchy model define a complete graph — or *clique* — provides the seminal point of departure for this investigation. Additional opportunities arise when the resulting graph is also directed and acyclic. Saaty (1986) recognized this: "Partially ordered sets with a finite number of elements can be conveniently represented by a directed graph".

Complete Directed Acyclic Graphs (DAGs) belong to a special class of graphs called *tournament graphs* (Moon (2015)). Nishizawa (1999) obliquely considered graphical methods for AHP consistency. The decision-theory literature however contains scant exploration of AHP's graphical characteristics and opportunities arising therefrom.

Entropy — a key concept from information theory — provides an abstract but rigorous framework for characterizing evidentiary strength (Cover & Thomas (2006)). In general, "the entropy of a graph is interpreted *as its structural information content* and serves as a complexity measure" (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Structural information content bears direct relevence to the present concern of evidentiary reasoning. Brown, et al (2020) studied the particular characteristics of tournament graphs.

What questions do entropies of AHP-based graphs allow us to answer? Endogenous to the AHP formulation itself, *relative entropy* quantifies the marginal informational contribution by a subset of elements within the evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive empirical data spanning of both positive and negative cases, as with Hué's approach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of *belief-function graphical models* to decompose their elements' relative evidentiary strength.

Evidentiary strength relates to the concept of *Value of Information* on which the disciplines of Information Economics (Birchler & Bütler (2007)) and Decision Analysis (Howard & Abbas (2016); Abbas & Hazen (2025)) are substantially based. The present investigation proposes an information-theoretic perspective. In contrast, information economics and decision analysis focus on utility values of information. Notwithstanding, Birchler & Bütler (2007) apply information-theoretic and probabilistic-state-space (along the lines of de Finetti (1974)) principles.

By what mechanism do we infer the differential information between hierarchically-adjacent AHP hierarchy models? Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has been well-researched (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost by ommission of elements from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for complete, mixed graphs with *weighted* edges. Brown, et al (2020) analyzed directed graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graphs comprised of weighted edges. The intensities of importance from the AHP comparative judgements provide graph-edge weights used here.

Second, relative entropies of hierarchically adjacent hierarchy models must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into contituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a complete graph — for each hierarchy model. Cover & Thomas (2006) describe *chain rules* for entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically coupled graphs.

Why is "heavy" machinery of graph entropy called-for? One might intiutively assume that individual comparative judgements suffice to inform us about the most-influential elements. The AHP's machinery however introduces "mutual coupling" between elements: "...many of the concepts derived for hierarchies also relate to systems with feedback" (Saaty (1986)). This calls into question the efficacy of sensitivity analyses based based on linear combinations of hierarchy-model priority vectors.

That comparative judgements define a graph — a network — provides the basis of their power. Our graph-entropy machinery accounts for the *aggegate effect* of all the evidence informing a decision or risk assessment. Individual importance intensities fail to provide this.

Decision Quality as a check on the completeness of evidentiary reasoning.

Decision Quality (DQ) represents a prominent applied decision-analysis framework that emphasizes evidentiary reasoning (Spetzler, Winter, Meyer (2016); Society of Decision Professionals (DQ)). Philosophically congruent with design-of-experiements practice (Cox & Reid (2000)), DQ provides a *de facto* standard of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. Introducing DQ here — independent of the mechanics of AHP — reinforces our logical completeness.

The pattern of reasoning underlying Saaty's (1986) AHP methodology loosely maps to the DQ framework. Figure 1 depicts the framework, comprised of six *links.*

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with prospcriptive judgements to provide analytical clarity. Saaty (1994) described this as "...the most creative part of decision making that has a significant effect on the outcome...".
- By definition, the act of deciding *requires* selection of one or more options from a discrete, precisely defined set of *Creative Alternatives.*
- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified. They specify

Figure 1: Decision-Quality framework. (From Society of Decision Professionals (DQ))

the decision makers' values, clarifying *why* the best alternative actually is the best.

- *Sound Reasoning* defines the logic employed in identifying the best alternative given the available evidence. Creighton (1909) cautioned that "...it is not possible to lay down definite rules for the guidance of thinking in every case." The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach, with accompanying advantages and disadvantages.

- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to act, a decision arguably has not occurred. The decision analysis was merely academic. Persistence with the status quo nearly always occurs among the alternatives under consideration. Such a selection should however result from applying sound reasoning to the relevant available evidence.

Now, the use of AHP by no means suffices to achieve DQ. Thoughtful application of AHP does adress a subset of DQ's links. Injudiciously applied, it does not however guarantee satisfaction of the von Neumann-Morgenstern axioms (Kochenderfer , 2015, p. 58) undergirding decision science. Featuring AHP as an approach to DQ in no way constitutes our objective here. We instead explore AHP's ability to highlight how subsets of decisional values and spans of evidence influence the results of sound reasoning.

Course of investigation.

We begin with an illustration of AHP methodology. A *real-world* case study — Titanium-alloy selection — demonstrates its application to preference-ranking a set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The complete graphs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement graphs associated with distinct hierarchy models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes weighted directed edges and unit-weight undirected edges. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted directed-graph Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based graph structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvil, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entroy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

AHP case study: Titanium-alloy selection for offshore pipelines.

Zeng, D., et al (2025) employed AHP to select the best candidate Titanium alloy from which to construct pipelines for use in offshore oil and gas extraction. Their analysis considers metallurgical technical details that are tangential to our purposes. Here, we principally focus on the AHP mechanics.

Zeng, D., et al (2025) provide a particularly useful case study for three reasons.

1. *It demonstrates a real-world application*. Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration. It involves nontrivial, idiosyncratic complexities that challenge our methodology.

2. *This case is recent*. Given that the AHP literature spans decades, one might succomb to the temptation to conclude that the method is anochronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically meaningful application.

3. *It strikes a balance between tractability and complexity*. Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demonstrate the use of relative entropy to couple hierarchically adjacent hierarchy models. Moreover, its results are straightforwardly reproducible.

The following offers neither criticism nor endorsement of Zeng, D., et al (2025). Their work merely serves to illustrate AHP mechanics.

We strive in the following to select AHP terminology that remains close to Saaty (1986, 1994). Variations in terminology occur througout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a "weight vector" or "weights". We employ *priority vector* here.

The following reiterates the case study in sufficient detail to reconstruct its results without explicit reference to Zeng, D., et al (2025). Beyond reproducing AHP mechanics, sufficient reference to its underlying principles is provided to justify assertions of complimentarity to the utility-optimization fundamentals of decision theory. We believe that the consolidated assembly of these concepts in itself constitutes a minor contribution in its own right.

Specify the evidentiary span.

Figure 2 reproduces the *hierarchy model* employed by Zeng, D., et al (2025) for their Titanium-alloy selection decision. It depicts a hierarchically structured decision. Their identified target hierarchy model $\mathfrak{H}_{TA_{\text{opt}}}$ describes a decisional activity: "Material selection of titanium alloy ⋯." Figure 2 identifies eight metallurgical elements, $B_1, \cdots, B_8$, and one economic one B_9. The latter suggests a basis for marginal utility.

The constituent criterion-to-alternative hierarchy models $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9} \subseteq \mathfrak{H}_{TA_{\text{opt}}}$ account for how each of three candidate alloys — TA_3, TA_{10}, TA_{36} — addresses the metallurgical desidirata $B_1, \cdots, B_9$. Zeng, D., et al (2025) describe an extensive material-testing regime to ascertain the the performance of each alloy with respect to the metallurgical criteria. Although comparative judgments introduce subjective aspects to the material-selection decision, Zeng, D., et al (2025) also bring to bear extensive empirical experimental evidence.

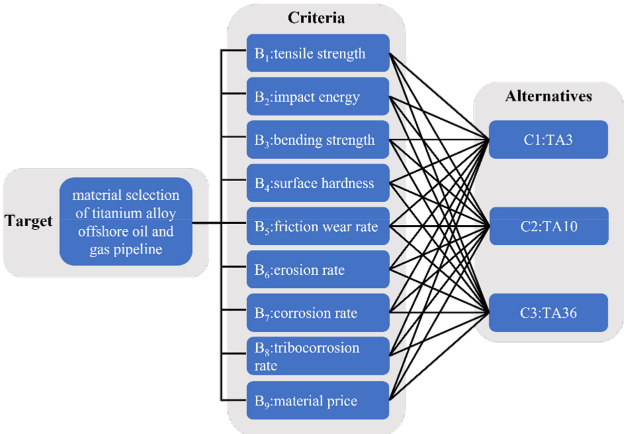
Visual, tree-graph depiction of the decision-element hierarchy aids in understanding by human decision-makers. Its idiosyncratic style notwithstanding, Figure 2 follows the ubiquitous pattern originating with Saaty (1986, 1994), and appearing throughout the MCDM literature. Structural elements of Figure 2 map to key DQ links from Figure 1.

- The rectangular box to the left "Target: material selection ⋯" at the left represents the *Goal or Focus*, in Saaty's (1986) terms. We use $\mathfrak{H}_{TA_{\text{opt}}}$ to denote this hierarchy model. A clear goal or focus results from thoughtful *Framing*, in DQ terms.

- The horizontally centered box identifies *Criteria*, denoted by $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9}$. Corresponding to DQ's *Values*, the these define the span of measurable criteria believed to materially distinguish between alternatives. Zeng, D., et al (2025) identified nine such criteria.

- Figure 2 explicitly depicts the *Alternatives* under consideration. That only three — TA_3, TA_{10}, and TA_{36}, each corresponding to a distinct Titanium alloy (Table 1 in Zeng, D., et al (2025)) — are considered makes their explicit representation here practical. When the number of alternatives is large — such as analyzing the risk of a large porfolio of distinct assets — showing all of them clutters the hierarch-model depiction.

Figure 2: Hierarchy model used for Titanium-alloy prioritization case study. (From Zeng, D., et al (2025), Figure 9)



That the target-to-criterion hierarchy model $\mathfrak{H}_{TA_{\text{opt}}}$ in Figure 2 spans nine elements $B_1, \cdots, B_9$ introduces a modicum of complexity. Saaty (1986) specifies a *transitive-consistency* criterion that, analogous to but stronger than the von Neumann-Morgenstern transitivity axiom (Kochenderfer , 2015, p. 58)

$$(A \succ B) \wedge (B \succ C) \Leftrightarrow A \succ C, \tag{1}$$

proscribes circular reasoning. Enforcing this constraint during the comparative-judgement process can prove challenging, particularly as the number of hierarchy-model elements grows.

Conventionally, AHP practitioners calculate a *Consistency Index.* This algebraically characterizes the degree to which both transitivity conditions are satisfied (Pant, et al (2022)). Representing comparative judgements as a directed graph provides an unambiguous mechanism to explicitly identify violations of von Neumann-Morgenstern transitivity.

Alignment of AHP to mainstream decision science involves specification of a utility function. AHP goals do not by default correspond to utilities in the conventional, decision-science sense. AHP practitioners can impose such an association exogenous to the mechanics of the methodolgy itself. Zeng, D., et al (2025) speak to qualities — values — of utility. They do not however explicitly articulate a utility function or theory.

Perform comparative judgements about elements' relative importances.

Apropos of DQ, AHP imbues values and tradeoffs into decision-making through comparative judgements: "...a matrix to carry out *pairwise comparisons* of the relative importance of the elements..." (Saaty , 1986, emphasis added). *Pairwise* is the operative term, here.

Practically, one assembles a list of unique, unordered element pairs for each hierarchy model. Explicit representation of hierarchy-element comparative judgements is frequently skipped in practice. The results are applied directly to a reciprocal matrix. Since comparative judgements themselves provide our excursion into graph-based evidentiary reasoning, we explicitly represent them here.

The logic of comparative judgement.

Consider a hierarchy model $\mathfrak{H}$ comprised of n elements, $\mathfrak{H} \supseteq \{e_1, \cdots, e_n\}$. Comparative judgement $\mathcal{J}_{\mathfrak{H}}$ entails $\binom{n}{2}$ distinct mappings

$$(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} > 0 \; \forall \, i \neq j, \tag{2a}$$

$$(e_j, e_i) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{j,i} = a_{i,j}^{-1}, \text{ and} \tag{2b}$$

$$(e_i, e_i) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,i} \equiv 1, \tag{2c}$$

where $a_{i,j}$ denotes the reciprocal-matrix element corresponding to (e_i, e_j). The mapping $\mathcal{J}_{\mathfrak{H}}$ imposes

$$\begin{array}{c} e_i \succcurlyeq e_j \Rightarrow a_{i,j} \geq 1 \\ e_i \prec e_j \Rightarrow a_{i,j} < 1 \end{array} . \tag{2d}$$

Consequently, after applying the comparative judgement $\mathcal{J}_{\mathfrak{H}}$, element ordering must be maintained in order to accurately interpret $a_{i,j}$. By (2a), $a_{i,j}$ is positive-definite. The $\binom{n}{2}$ distinct mappings exclude the reciprocal-matrix diagonal elements $a_{i,i}$, which by (2c) are axiomatically assigned values of unity.

Comparative judgements for Titanium-alloy prioritization.

Figure 2 contains ten hierarchic models: One target-to-criterion model comprised of nine elements; and nine criterion-to-alternative models, each with three elements. Consequently, $\binom{9}{2} = 36$ element pairs must be considered for the target-to-criterion model. Each criterion-to-alternative model entails $\binom{3}{2} = 3$ element pairs. Each element pair corresponds to an edge for the graph by which the hierarchy model is represented.

Table 1a contains the comparative judgements assigned by Zeng, D., et al (2025) for each element pair depicted in the target-to-criterion hierarchy model in Figure 2. Table 1b contains element-pair judgements for each of the criterion-to-alternative models. Explicit depiction of the comparative judgements here contributes to the clarity of the present course of reasoning. In particular, the edge lists in Tables 1a and 1b provide a key of departure for subsequent graphical extension of the AHP methodology.

Table 1a: Comparative judgements for each element pair within the target-to-criterion hierarchy model $\mathfrak{H}_{TA_{\text{opt}}}$ of Figure 2. (After Zeng, D., et al (2025), Table 10.)

reference element $e_{\mathfrak{H}_i}$	comparative element $e_{\mathfrak{H}_j}$	intensity of importance $\mathcal{J}_{\mathfrak{H}}$
B_1	B_2	4
B_1	B_3	2
B_1	B_5	4
B_1	B_6	2
B_1	B_7	2
B_1	B_9	2
B_2	B_4	3
B_2	B_6	1
B_2	B_9	1
B_3	B_2	3
B_3	B_4	3
B_3	B_5	2
B_3	B_6	2
B_3	B_7	2
B_3	B_9	3
B_5	B_2	3
B_5	B_4	3
B_5	B_6	1
B_5	B_7	2
B_5	B_9	2
B_6	B_4	2
B_6	B_7	5
B_6	B_9	2
B_7	B_2	3
B_7	B_4	3
B_7	B_9	2
B_8	B_1	5
B_8	B_2	5
B_8	B_3	5
B_8	B_4	5
B_8	B_5	5
B_8	B_6	5
B_8	B_7	5
B_8	B_9	9
B_9	B_4	1

Table 1b: Comparative judgements for each element pair within each of the criterion-to-alternative hierarchy models $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{b_9}$ in Figure 2. (After Zeng, D., et al (2025), Table 8).

hierarchy model $\mathfrak{H}$	reference element $e_{\mathfrak{H}_i}$	comparative element $e_{\mathfrak{H}_j}$	intensity of importance $\mathcal{J}_{\mathfrak{H}}$
B_1	TA_3	TA_{10}	3
B_1	TA_{36}	TA_3	3
B_1	TA_{36}	TA_{10}	5
B_2	TA_{10}	TA_3	3
B_2	TA_{36}	TA_3	5
B_2	TA_{36}	TA_{10}	3
B_3	TA_3	TA_{10}	3
B_3	TA_{36}	TA_3	3
B_3	TA_{36}	TA_{10}	5
B_4	TA_3	TA_{10}	2
B_4	TA_{36}	TA_3	3
B_4	TA_{36}	TA_{10}	3
B_5	TA_3	TA_{10}	3
B_5	TA_{36}	TA_3	3
B_5	TA_{36}	TA_{10}	7
B_6	TA_3	TA_{10}	2
B_6	TA_{36}	TA_3	3
B_6	TA_{36}	TA_{10}	3
B_7	TA_3	TA_{10}	3
B_7	TA_3	TA_{36}	5
B_7	TA_{10}	TA_{36}	3
B_8	TA_3	TA_{10}	2
B_8	TA_{36}	TA_3	3
B_8	TA_{36}	TA_{10}	5
B_9	TA_3	TA_{10}	3
B_9	TA_{36}	TA_3	3
B_9	TA_{36}	TA_{10}	5

Assigning intensities of importance.

Now, Saaty (1986) Table 1 and Saaty (1994) Table 1 offer a "fundamental scale" containing heuristics for assignment of *intensity-of-importance* values to hierarchy-model element pairs. Saaty (2001b) offered a *psychophysics* justification for this scale, and recommended $a_{i,j} \in [1/9, 9]\ \forall (i,j)$ when assigning values to importance intensities. Zeng, D., et al (2025) employ this scale (their Table 7), as do many other AHP applications throughout the MCDM literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains no explicit prescription for this scale for importance-intensity assignment.

Saaty (1986, 2001b) do however speak to preferential ordering, transitive consistency, and homogeneity considerations. Consequently, alternative, more-rigorous importance-intensity assignments can be considered. Application of the personal-probability indifference principle arises as one option (Savage (1972); de Finetti (1974)). In fact, Saaty (2001a) grounds AHP logic in *Perron-Frobenius Theory*, which begins from state-transition probabilities (Horn & Johnson (2012)). Alternatively, Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-emprical approaches to *measurement*. Application of such alternatives in lieu of Saaty's "fundamental scale" deflects a common critique of AHP's logical rigor by recasting comparative judgments in terms of probabilism (Pettigrew (2016)).

Constructing comparative-judgement tables.

The construction of Tables 1a and 1b anticipates their use as edge lists for comparative-judgement graphs. The AHP literature — Saaty (1986, 1994, 2003), in particular — prescribes nothing about the ordering of element pairs in comparative judgement. Construction of Tables 1a and 1b follows the following convention.

- Populate the "intensity of importance" column with the value greater than or equal to unity $a_{i,j} \mid (e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} \geq 1$ for each element pair (e_i, e_j). If $(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} < 1$, then pick $(e_j, e_i) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{j,i} \geq 1$ instead.
- Assign to the "reference element" column the corresponding most-preferred element $e_j \mid e_j \succcurlyeq e_i$, associated with the preferred intensity-of-importance value $a_{i,j}$. This element subsequently corresponds to the head of the directed-graph edge connecting the two elements.
- Populate the "comparative element" column with the less-preferred element $e_i \mid e_i \prec e_j$. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences into the procedure. First, the structure of the table endogenously contains the directive structure of the graph. This simplifies the algorithmic logic used to construct the resultant complete graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage (1972); de Finetti (1974)) in lieu of the less-widely-recognized "fundamental scale".

Comparative-judgement indifference and its consequences.

That Table 1a contains unit-valued importance intensities — occurrences of $(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto 1 \mid i \neq j$ — raises both interest and a challenge. Saaty's (1986) "fundamental scale" explicitly allows for this. Such occurrences represent undirected edges in an otherwise-directed graph. Four undirected edges appear in Table 1a. Consequently, the comparative judgements $\mathcal{J}_{TA_{\text{opt}}}$ for $\mathfrak{H}_{TA_{\text{opt}}}$ are represented by a *mixed graph,* containing both directed and undirected edges.

We subsequently handle this in construction of the graph Laplacians, on which entropy calculations are based. Our Laplacians must account for both directed and undirected edges, as well as other-than-unity edge weights. Alluded to previously, graph-Laplacian constructions for distinct graph classes appear separately in the literature (Wong & You (1985); Biggs (2012); Godsil & Royle (2001); Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012); Brown, et al (2020); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Judiciously combining graph-class-specific Laplacian structures with our mixed-graph scenario represents a key task of the present investigation.

Transform comparative judgments into *priority vectors*.

AHP *priority vectors* provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructing a *reciprocal matrix* from the comparative judgements for each hierarchy model. Reciprocal matrices' values are assigned according to (2a) - (2c).

Construction of reciprocal matrices from comparative judgements.

As the name implies, a reciprocal matrix is a matrix of comparative judgements within which upper-diagonal elements are reciprocals of the lower. Consider again a hierarchy model $\mathfrak{H}$ comprised of n elements, $\mathfrak{H} \supseteq \{e_1, \cdots, e_n\}$.

Then the corresponding reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$ becomes

$$\boldsymbol{A}_{\mathfrak{H}} = \begin{bmatrix} 1 & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & 1 & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & 1 \end{bmatrix} \tag{3a}$$

where values of $a_{i,j}$ are assigned according to $\mathcal{J}_{\mathfrak{H}}$ as specified by (2a) - (2c).

Illustratively, the reciprocal-matrix for criterion-to-alternative hierarchy model $\mathfrak{H}_{B_1}$ becomes

$$A_{B_1} = \begin{pmatrix} 1 & 3 & 1/3 \\ 1/3 & 1 & 1/5 \\ 3 & 5 & 1 \end{pmatrix}, \tag{3b}$$

where Table 1b specifies matrix-element values. Zeng, D., et al (2025) exhaustively represent the reciprocal-matrix set $\boldsymbol{A}_{TA_{\text{opt}}}$ and $\left\{\boldsymbol{A}_{B_1}, \cdots, \boldsymbol{A}_{B_9}\right\}$ in their Table 10 and equation (13), respectively. Complete reproduction here is unnecessary given the pattern (3a).

Priority-vector calculation.

The AHP-application literature contains different approaches to translating a reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$ for hierarchy model $\mathfrak{H}$ into a priority vector $\boldsymbol{\omega}_{\mathfrak{H}}$. We summarize here comparisons previously described by Saaty & Vargas (1984) and Saaty (1990).

First, that $\boldsymbol{\omega}_{\mathfrak{H}}$ is a projection onto the unit simplex Δ^n, where

$$\Delta^n \triangleq \left\{ \boldsymbol{x} \in \mathbb{R}^n \middle| \left(\sum_i x_i = 1 \right) \wedge (x_i \geq 0\ \forall i) \right\}, \tag{4}$$

is highly significant. Both normalized measurements of hierarchy-model elements (11a) and strengths of preference between alternatives (12b) are constrained to the interval $[0, 1]$. That $\boldsymbol{\omega}_{\mathfrak{H}} \in \Delta^n$ ensures the latter given the former.

Summary of priority-vector approaches.

Within the AHP literature, priority-vector calculation preponderantly occurs using one of two approaches. Saaty (2001a, 2003) preferred an eigenvector-based approach. Researchers often find a geometric-mean approach more-convenient. We also briefly consider a more-empirical approach to obtaining priority vectors.

Principal-eigenvector method. Saaty (2001a, 2003) strongly argued for the principal eigenvector as the definitive approach to calculating priority vectors $\boldsymbol{\omega}_{\mathfrak{H}}$. Consider the eigenvector/eigenvalue decomposition

$$\boldsymbol{A}_{\mathfrak{H}} \boldsymbol{\Omega}_{\mathfrak{H}} = \boldsymbol{\Omega}_{\mathfrak{H}} \boldsymbol{\Lambda}_{\mathfrak{H}} \tag{5a}$$

of the reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$ for a hierarchy model $\mathfrak{H}$. As usual, $\boldsymbol{\Omega}_{\mathfrak{H}}$ denotes a matrix whose columns are the eigenvectors $\boldsymbol{A}_{\mathfrak{H}}$. $\boldsymbol{\Lambda}_{\mathfrak{H}}$ is a diagonal matrix of the corresponding eigenvalues and $\boldsymbol{\lambda}_{\mathfrak{H}}$ a vector of the diagonal elements of $\boldsymbol{\Lambda}_{\mathfrak{H}}$. The resulting principal eigenvalue then becomes

$$\lambda_{\mathfrak{H}}^{\mathcal{P}} = [\boldsymbol{\lambda}_{\mathfrak{H}}]_{\text{argmax}(\boldsymbol{\lambda}_{\mathfrak{H}})}\,. \tag{5b}$$

The priority vector $\boldsymbol{\omega}_{\mathfrak{H}}$ follows from the corresponding principal eigenvector

$$\left[\boldsymbol{\omega}_{\mathfrak{H}}^{\mathcal{P}}\right]_j = \frac{[\boldsymbol{\Omega}_{\mathfrak{H}}]_{j,\text{argmax}(\boldsymbol{\lambda}_{\mathfrak{H}})}}{\sum_i [\boldsymbol{\Omega}_{\mathfrak{H}}]_{i,\text{argmax}(\boldsymbol{\lambda}_{\mathfrak{H}})}}. \tag{5c}$$

Saaty (2001a) grounded the rationale for this priority-vector choice in the *Perron-Frobenius Theorem* (Horn & Johnson (2012); Hawkins (2021)). He suggested that priority-ordering stability by AHP proves highly sensitive to the transitive consistency of comparative judgements. More-strenuous than von Neumann-Morgenstern preferential-ordering transitivity (1), *AHP transitivity* requires

$$a_{i,j} a_{j,k} = a_{i,k}, \tag{6a}$$

also applying to intensities of importance.

Consider a stylized illustration of the consequence of (6a) based on preferences for pizza. Say that one prefers margherita-style pizza twice as much as pepperoni pizza. Also, pepperoni is preferred three times as much as ham-and-pineapple pizza. Reciprocal-matrix notation represents this as

$$\begin{array}{c} a_{\text{margherita,pepperoni}} = 2 \\ \text{and} \\ a_{\text{pepperoni,ham-pinapple}} = 3 \end{array}.$$

The AHP-transitivity condition (6a) consequently requires

$$a_{\text{margherita,ham-pineapple}} = 6.$$

The von Neumann-Morgenstern transitivity axiom (1) certainly requires the preferential ordering. The necessity of intensity-of-importance transitivity poses a more-complicated question. When probabilistically framed, the validity of (6a) depends on statistical independence, among other conditions.

By induction, (6a) extends to higher orders

$$a_{1,2} a_{2,3} \cdots a_{n-1,n} = a_{1,n}. \tag{6b}$$

In fact, Saaty (2001a) proves a limiting case for transitively consistent reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$

$$\lim_{k \to \infty} \frac{\boldsymbol{A}_{\mathfrak{H}}^k \boldsymbol{\omega}}{\boldsymbol{\omega}^T \boldsymbol{A}_{\mathfrak{H}}^k \boldsymbol{\omega}} \to \boldsymbol{\omega}_{\mathfrak{H}}^{\mathcal{P}} \tag{6c}$$

given some arbitrary positive-definite $\boldsymbol{\omega}$. This associates $\boldsymbol{\omega}_{\mathfrak{H}}$ with the limiting case of transitive consistency specified by (6a).

Geometric-mean methods. Forman & Peniwati (1996), Csató (2019), and others argued that arithmetic or geometric means provided suitable priority vectors in some cases. Zeng, D., et al (2025) employed a geometric-mean approach

$$\left[\boldsymbol{\omega}^{\mathcal{G}}_{\mathfrak{H}}\right]_i = \frac{\left(\prod_{j=1}^{n} a_{i,j}\right)^{\frac{1}{n}}}{\sum_{i=1}^{n}\left(\prod_{j=1}^{n} a_{i,j}\right)^{\frac{1}{n}}}. \tag{7}$$

Saaty (1990, 2001a) however insinuated an accompanying risk of rank-reversal with this approach even under near-trivial conditions. Nonetheless, when a high degree of comparative-judgement transitivity (6a) occurs, results from (7) converge to those produced by (5c). The approach (7) offers transparency and reproducibility advantages that many researchers find more-straightforward than eigenvector calculation.

Entropy-weight method. Now, the priority-vector methods of (5c) and (7) are solely based reciprocal matrices constructed from subjective comparative judgements. Sometimes more-empirical bases for priority vectors prove useful. This allows for consideration of phenomenological information exogenous to comparative judgements.

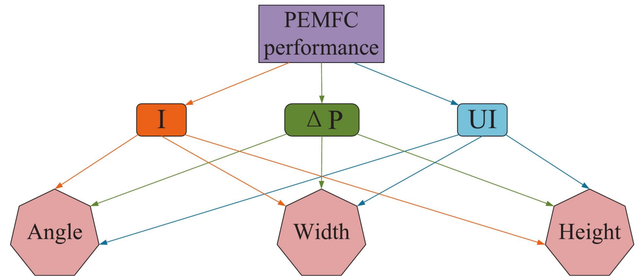
The *Entropy-Weight Method* provides such an approach. Examples of its use appear in Qu, et al (2022); Shen & Wu (2023); Ren & Sun (2025). It requires a larger number of alternatives than Zeng, D., et al (2025) considered. Practically, one normalizes — standardizes to a scale in [0, 1] along the lines of (11a) — the measurements to which priority vectors are applied. A series of calculations including entropy leads to an empirically-based priority vector.

Bu, et al (2026) applied such a hybrid approach to the design of fuel cells. Figure 3 reproduces their hierarchy model. The intermediate criteria — current density I, pressure drop ΔP, Oxygen-uniformity index and UI — represent device physical-operating characteristics. Represented as heptagons, sub-criteria $Angle$, $Width$, and $Height$ correspond to geometric design parameters.

Bu, et al (2026) employed a physics-based computational model to establish relationships between the device-geometry elements and the physical operating parameters. These simulation results led to an empirical basis for an entropy-based priority vector. Combining this with a priority vector based on comparative judgements produced a "hybrid" priority vector on which their design decision was based.

Now, entropy-weight priority vectors are not necessarily subject to constraints associated with transitive

Figure 3: Hierarchy model used in design of Proton-Exchange Membrane Fuel Cell (PEMFC). (From Bu, et al (2026) Figure 5)



consistency along the lines of (6a)-(6c). Nonetheless, exemplified by Bu, et al (2026), empirical phenomenologically based evidence can prove more-informative than subjective comparative judgements alone. The highest-quality decision depends on judicious use of the most-informative evidence available.

Priority vectors used for Titanium-alloy prioritization. Table 2 lists the priority-vector elements for each hierarchy model depicted by Figure 2 based on the comparative judgements in Tables 1a and 1b. The priority vector $\boldsymbol{\omega}^{\mathcal{P}}_{\mathfrak{H}}$ values result from (5c). The *linalg.eig* method from the python *numpy* library (Harris, et al (2020)) produced these results. The "$\mathbb{L}^2$ distance" column contains the Euclidean $\mathbb{L}^2$ distance between $\boldsymbol{\omega}^{\mathcal{P}}_{\mathfrak{H}}$ and $\boldsymbol{\omega}^{\mathcal{G}}_{\mathfrak{H}}$ for each. The TA_{opt} hierarchy model corresponds to the target-to-criterion model in Figure 2.

Visual comparison between Table 2 here and Figure 25 and Table 12 from Zeng, D., et al (2025) reveals that priority-vector values for hierarchy models $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9}$ coincide exactly. The "$\mathbb{L}^2$ distance" column in the former indicates negligible differences between $\boldsymbol{\omega}^{\mathcal{P}}_{B_i}$ and $\boldsymbol{\omega}^{\mathcal{G}}_{B_i}$. The arguments by Forman & Peniwati (1996) and Csató (2019) apparently hold up in this case.

The priority-vector values for the TA_{opt} hierarchy model reported by Table 11 from Zeng, D., et al (2025) differ from those in Table 2 beginning at about the fifth significant digit. This is consistent with the "$\mathbb{L}^2$ distance" column. As the hierarchy-model element span expands, achieving transitive consistency — according to both (1) and (6a) — becomes increasingly challenging. Commensurately, the equivalence between (5c) and (7) becomes increasingly tenuous.

For large sets of alternatives, the divergence between $\boldsymbol{\omega}^{\mathcal{P}}_{\mathfrak{H}}$ and $\boldsymbol{\omega}^{\mathcal{G}}_{\mathfrak{H}}$ increases the likelihood of rank reversal when using the latter. Zeng, D., et al (2025) only considered three alternatives. Provided that criterion-to-alternative priority vectors $\boldsymbol{\omega}_{B_1}, \cdots, \boldsymbol{\omega}_{B_9}$ and input measurements to which they are applied are sufficiently distinct, the risks of rank reversal are small.

Table 2: Priority vectors for pipeline-material selection case study Zeng, D., et al (2025) based on comparative judgements in Tables 1a and 1b.

hierarchy model $\mathfrak{H}$	element $e_{\mathfrak{H}_i}$	priority vector $\left[\boldsymbol{\omega}^{\mathcal{P}}_{\mathfrak{H}}\right]_i$	$\mathbb{L}^2$ distance $\left\lVert\boldsymbol{\omega}^{\mathcal{P}}_{\mathfrak{H}} - \boldsymbol{\omega}^{\mathcal{G}}_{\mathfrak{H}}\right\rVert$
B_1	TA_3	0.2582	2.850E-32
B_1	TA_{10}	0.1047	
B_1	TA_{36}	0.6370	
B_2	TA_3	0.1047	1.560E-32
B_2	TA_{10}	0.2582	
B_2	TA_{36}	0.6370	
B_3	TA_3	0.2582	2.850E-32
B_3	TA_{10}	0.1047	
B_3	TA_{36}	0.6370	
B_4	TA_3	0.2493	2.003E-32
B_4	TA_{10}	0.1571	
B_4	TA_{36}	0.5936	
B_5	TA_3	0.2426	1.329E-32
B_5	TA_{10}	0.0879	
B_5	TA_{36}	0.6694	
B_6	TA_3	0.2493	2.003E-32
B_6	TA_{10}	0.1571	
B_6	TA_{36}	0.5936	
B_7	TA_3	0.6370	2.489E-32
B_7	TA_{10}	0.2583	
B_7	TA_{36}	0.1047	
B_8	TA_3	0.2300	7.704E-34
B_8	TA_{10}	0.1220	
B_8	TA_{36}	0.6483	
B_9	TA_3	0.2583	2.850E-32
B_9	TA_{10}	0.1047	
B_9	TA_{36}	0.6370	
TA_{opt}	B_1	0.1436	5.473E-05
TA_{opt}	B_2	0.0462	
TA_{opt}	B_3	0.1156	
TA_{opt}	B_4	0.0353	
TA_{opt}	B_5	0.0876	
TA_{opt}	B_6	0.0858	
TA_{opt}	B_7	0.0711	
TA_{opt}	B_8	0.3793	
TA_{opt}	B_9	0.0361	

Evaluate comparative judgements for *transitive* consistency.

Glenn Shafer once opined, "...probability is not really about numbers; it is about the structure of reasoning..." (quoted in Pearl (1988), p. 15). The AHP methodology contains mechanisms to identify inconsistencies and contradictions in the structure of reasoning (Saaty (2003); Pant, et al (2022)). Here, we introduce *consistency indices*, AHP's out-of-the-box mechanism to quantify the degree of comparative judgements' consistency in terms of (6a). We also demonstrate a graph-theoretic approaches to identify von Neumann-Morgenstern transitivity violations.

Although this analysis could occur as part of the comparative-judgement stage, it typically appears in MCDM literature after priority-vector calculation. The consistency indices reuse intermediate results of the latter when employing the principal-eigenvector approach (5c).

Saaty (1986) presents an extensive set of definitions, axioms, and theorems leading to

$$\lambda^{\mathcal{P}}_{\mathfrak{H}} \to n, \tag{8}$$

where n denotes the order of *perfectly consistent* reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$ and $\lambda^{\mathcal{P}}_{\mathfrak{H}}$ is defined in (5b). Being perfectly consistent, $\boldsymbol{A}_{\mathfrak{H}}$ invariantly satisfies (6a)-(6c).

Transitive consistency bears important decision-theoretic consequences: "...when alternatives are scaled through paired comparisons, adding a new alternative can change the relative ranking of the old ones when the judgements are inconsistent or when several criteria are used" (Saaty (1986)).

A consistency index for reciprocal matrices.

A consequence of (8),

$$CI = \frac{\lambda^{\mathcal{P}}_{\mathfrak{H}} - n}{n - 1} \tag{9a}$$

provides a commonly used *Consistency Index* for reciprocal matrix $\boldsymbol{A}_{\mathfrak{H}}$ (Pant, et al (2022)). From (8), one obtains $CI \to 0$ when $\boldsymbol{A}_{\mathfrak{H}}$ is perfectly consistent in terms of (6a).

A *Consistency Ratio*

$$CR = \frac{CI}{RI} \tag{9b}$$

accompanies CI. In (9b), the denominator RI results from calculating CI for a large number of reciprocal matrices with randomly generated comparative judgements. Tabulated RI values are commonly repeated. Reproducing the common result with atypically greater numerical precision, Donhegan & Dodd (1991) reported $RI_3 = 0.4887$ for reciprocal-matrix order $n = 3$ and

$RI_9 = 1.3733$ for $n = 9$. Use of these values leads to small disparities between calculations reproduced here and those reported by Zeng, D., et al (2025).

Consistency-index values for the material-selection case study.

Table 3 lists CI (9a), CR (9b), $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ (5b), and the order of $\boldsymbol{A}_{\mathfrak{H}}$ for each of the ten hierarchy models defined by Figure 2. As with the priority vectors, the values for $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ coincide almost exactly with those in Table 12 from Zeng, D., et al (2025), which actually reports $\boldsymbol{\omega}_{\mathfrak{H}}^{\mathcal{G}}$. That CI and CR employ RI values based Donhegan & Dodd (1991) — which contain greater numerical precision — contributes to the small disparities. Comparable disparities occur for the target model TA_{opt} between Table 3 values and those from Zeng, D., et al (2025) Table 11.

Table 3: Consistency indices for hierarchy models defined by Figure 2 from the offshore-pipeline material-selection case study (Zeng, D., et al (2025)).

hierarchy model $\mathfrak{H}$	CI (9a)	CR (9b)	$\lambda_{\mathfrak{H}}^{\mathcal{P}}$ (5b)	order of $\boldsymbol{A}_{\mathfrak{H}}$
B_1	0.0193	0.039	3.039	3
B_2	0.0193	0.039	3.039	3
B_3	0.0193	0.039	3.039	3
B_4	0.0268	0.055	3.054	3
B_5	0.0035	0.007	3.007	3
B_6	0.0268	0.055	3.054	3
B_7	0.0192	0.039	3.039	3
B_8	0.0019	0.004	3.004	3
B_9	0.0193	0.040	3.039	3
TA_{opt}	0.0702	0.051	9.562	9

The coincidence between $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ and n the order of $\boldsymbol{A}_{\mathfrak{H}}$ in Table 3 provides the raw indication of the extent to which (8) is satisfied. Except for the $\mathfrak{H}_{TA_{\text{opt}}}$, the quantities coincide very closely. Pant, et al (2022) reiterate the heuristic $CR \lesssim 0.1$ as a "good-enough" condition for most purposes. All of the CR values in Table 3 meet this criterion.

Remediation of reciprocal-matrix transitive inconsistencies.

Saaty (2003) describes a procedure to "...transform a positive reciprocal matrix to a near-consistent matrix." This involves applying perturbations to the matrix $\boldsymbol{A}_{\mathfrak{H}}$ following a procedure described by Harker (1987). To be clear, the transform enforces importance-intensity transitivity (6a) as well as von Neumann-Morgenstern transitivity (1). Performing such a procedure may lead to an epistemic conundrum: One applies algorithmic logic to override comparative judgements.

Saaty (2003) elsewhere argued, "When we deal with tangibles, a pairwise comparison judgment matrix may be perfectly consistent but irrelevant and far off the mark of the true values. For several reasons a modicum of inconsistency may be considered as a good thing and forced consistency without knowledge of the precise values as an undesirable compulsion." Our pizza-preference thought exercise above illustrates.

Graphical detection of transitive inconsistencies.

The graphical structure of pairwise judgements motivates an unambiguous approach to detect violations of von Neumann-Morgenstern transitivity (1). Consider the comparative-judgement graph $\mathcal{G}$ for hierarchy model $\mathfrak{H}$. The preceding subsection on comparative-judgement tables described how comparative judgements are represented as a list of directed graph edges for a given $\mathfrak{H}$. So then, violations of (1) become straightforwardly detectable as cycles in $\mathcal{G}$.

Previously described comparative-judgement indifferences complicate this approach modestly. Specifically, edges associated with unit-valued intensities of importance are undirected edges. Their presence can introduce cycles whose presence masks violations of (1) associated with directed edges. We must therefore apply our directed-graph acyclicity test above to a spanning subgraph $\check{\mathcal{G}}$. By definition, a spanning subgraph $\check{\mathcal{G}}$ of $\mathcal{G}$ contains all of the nodes of the latter, but omits one or more edges (Godsil & Royle (2001)). In our case, $\check{\mathcal{G}}$ omits from $\mathcal{G}$ edges with unit-valued intensities of importance.

Commonly available open-source software libraries contain off-the-shelf logic to detect cycles in directed graphs. Illustratively, the python library *networkx* contains a built-in method $is\_directed\_acyclic\_graph$, which returns a value of $True$ or $False$ depending on whether or not the directed graph $\mathcal{G}$ contains cycles (Hagberg, Schult, Swart (2008)). The library also methods to explicitly identify directed-graph cycles if they are present. Such feedback to the comparative-judgement panel serves to sharpen their reasoning about hierarchy-model elements' relative importance. Applying this method to the spanning subgraph $\check{\mathcal{G}}_{TA_{\text{opt}}}$ based on $\mathfrak{H}_{TA_{\text{opt}}}$ reveals no transitivity violations within the directed edges. The complete graph $\mathcal{G}_{TA_{\text{opt}}}$ does however contain cycles, indicating transitivity violations.

Prioritize alternatives based on comparative judgements and measurements of hierarchy-model elements.

We now turn our attention to prioritization of alternatives by the AHP. We first describe the mechanics for the general case. We then apply the machinery to the Titanium-alloy selection case study.

The mechanics of prioritizing alternatives.

Prioritization of alternatives by the AHP follows a linear operation. The following discussion employs a linear-algebra notation in lieu of commonly used nested summations. This offers conceptual-clarity advantages. We develop the notation for two "layers" of hierarchy models, $\mathfrak{H}_{\text{target}} \supseteq \{\mathfrak{H}_1, \cdots, \mathfrak{H}_M\}$. Induction extends the notation to more-complex hierarchical structures.

Measurement-matrix construction. The measurement matrix contains the enumerated evidence weighed during multi-criterion decision making. Consider hierarchy model $\mathfrak{H}_n$ spanning N elements $\mathfrak{H}_n \supseteq \left\{ e_{\mathfrak{H}_{n,1}}, \cdots, e_{\mathfrak{H}_{n,N}} \right\}$. Expressed in matrix form, the measurement matrix for all hierarchy models within the span $\mathfrak{H}_{\text{target}} \supseteq \{\mathfrak{H}_1, \cdots, \mathfrak{H}_N\}$ appears as

$$\boldsymbol{R}_{\mathfrak{H}} = \left[\begin{array}{cccc} \boldsymbol{r}_{\mathfrak{H}_1} & \boldsymbol{r}_{\mathfrak{H}_2} & \cdots & \boldsymbol{r}_{\mathfrak{H}_N} \end{array} \right], \tag{10a}$$

where

$$\boldsymbol{r}^T_{\mathfrak{H}_n} = \left[r_{\mathfrak{H}_{n,1}}, \cdots, r_{\mathfrak{H}_{n,M}} \right] \tag{10b}$$

contains measurements for each of M alternatives with respect to hierarchy-model element $e_{\mathfrak{H}_n}$.

The constituent elements for each hierarchy model $\mathfrak{H}_n$ must be treated as distinct entities. Measurements of elements $e_{\mathfrak{H}_{m,m'}}$ and $e_{\mathfrak{H}_{n,n'}}$ are plausibly — thought not necessarily in practice — independent. The elements $\{e_{Angle}, e_{Height}, e_{Width}\}$ in Figure 3 exemplify an exception. Such occurrences are however rare in practice.

Element-measurement normalization. Our priority vectors reside on the unit simplex Δ^n (4). In order to constrain the *strengths of preference* among the alternatives to $[0, 1]$, we similarly normalize the element-measurement vectors.

Mechanics of element-measurement normalization. Normalization maps the elements of our hierarchy-model element-measurement vectors $\boldsymbol{r}_{\mathfrak{H}_n}$ to the interval $[0, 1]$:

$$\boldsymbol{r}_{\mathfrak{H}_n} : \mathcal{P}_{e_n} \mapsto \boldsymbol{p}_{\mathfrak{H}_n} \mid p_{\mathfrak{H}_{n,n'}} \in [0, 1] \ \forall \, n'. \tag{11a}$$

The measurment-normalization map $\mathcal{P}_{e_n}$ is defined by

$$p_{\mathfrak{H}_{n,n'}} = \begin{cases} \frac{r_{\mathfrak{H}_{n,n'}} - \min\{\boldsymbol{r}_{\mathfrak{H}_n}\}}{\max\left\{\boldsymbol{r}_{\mathfrak{H}_n}\right\} - \min\{\boldsymbol{r}_{\mathfrak{H}_n}\}} & if \ e_{\mathfrak{H}_n} \text{ is beneficial} \\ \frac{\max\left\{\boldsymbol{r}_{\mathfrak{H}_n}\right\} - r_{\mathfrak{H}_{n,n'}}}{\max\left\{\boldsymbol{r}_{\mathfrak{H}_n}\right\} - \min\{\boldsymbol{r}_{\mathfrak{H}_n}\}} & if \ e_{\mathfrak{H}_n} \text{ is deleterious} \end{cases}. \tag{11b}$$

We seek in all cases the following element-measurement normalization behavior:

$$\max_{n'} \left\{ \boldsymbol{p}_{\mathfrak{H}_{n,n'}} \right\} = 1 \tag{11c}$$

and

$$\min_{n'} \left\{ \boldsymbol{p}_{\mathfrak{H}_{n,n'}} \right\} = 0. \tag{11d}$$

The resulting normalized element-measurement matrix now becomes

$$\boldsymbol{P}_{\mathfrak{H}} = \left[\begin{array}{cccc} \boldsymbol{p}_{\mathfrak{H}_1} & \boldsymbol{p}_{\mathfrak{H}_2} & \cdots & \boldsymbol{p}_{\mathfrak{H}_M} \end{array} \right]. \tag{11e}$$

The *beneficial* versus *deleterious* characterizations in (11b) indicate whether increasing the measured value of element $e_{\mathfrak{H}_n}$ increases or decreases, respectively, the degree to to which a given alternative is preferred. Illustrating with a utility analogy, if $e_{\mathfrak{H}_n}$ is *beneficial*, then increasing the value of $r_{\mathfrak{H}_{n,n'}}$ returns greater utility of $\mathfrak{H}_{\text{target}}$, all other things being equal. In contrast, if $e_{\mathfrak{H}_n}$ is deleterious, then increasing $r_{\mathfrak{H}_{n,n'}}$ diminishes the utility of $\mathfrak{H}_{\text{target}}$.

Element-mesasurement normalization and rank reversal. The operation (11b) provides a key mechanism by which rank reversal can occur, regarding which Saaty (1986, 1994) warned. Consider a scenario in which a new alternative is introduced for consideration. If one or more of that new alternative's measurements $r_{\mathfrak{H}_{n,\ell}}$ fall outside of the range of prior alternatives — that is, $r_{\mathfrak{H}_{n,\ell}} \notin \left[\min_n \{\boldsymbol{r}_{\mathfrak{H}_n}\}, \max_n \left\{ \boldsymbol{r}_{\mathfrak{H}_n} \right\} \right]$ — then (11b) leads to different normalized measurements in $\boldsymbol{P}_{\mathfrak{H}}$ for all previously evaluated alternatives. This reëvaluation certainly alters the strengths of preference for each previously evaluated alternative. Consequently, they also provide a likely rank-reversal mechanism.

Prioritization of alternatives. The preferential ranking of alternatives results from a linear operation applying priority vectors to our normalized hierarchy-element measurement matrix $\boldsymbol{P}_{\mathfrak{H}}$. Representing the priority vectors associated with $\{\mathfrak{H}_1, \cdots, \mathfrak{H}_M\}$ in matrix form produces

$$\Omega_{\mathfrak{H}} = \left[\begin{array}{cccc} \boldsymbol{\omega}_{\mathfrak{H}_1} & \boldsymbol{\omega}_{\mathfrak{H}_2} & \cdots & \boldsymbol{\omega}_{\mathfrak{H}_M} \end{array} \right]. \tag{12a}$$

The *strength of preference* among the alternatives finally becomes

$$\boldsymbol{p}_{\text{target}} = \left(\boldsymbol{P}_{\mathfrak{H}} \otimes \Omega_{\mathfrak{H}} \right) \boldsymbol{\omega}_{\text{target}}, \tag{12b}$$

where $\boldsymbol{\omega}_{\text{target}}$ denotes the priority vector associated with hierarchy model $\mathfrak{H}_{\text{target}}$ and the Kronecker product is defined as (Graham (2018))

$$[\boldsymbol{A} \otimes \boldsymbol{B}]_{i,j} = a_{i,j} b_{i,j}.$$

What does $\boldsymbol{p}_{\text{target}}$ represent? The the elements of the vector $\boldsymbol{p}_{\text{target}}$ enumerate the *strength of preference* for each corresponding alternative within the range considered by the multi-criterion decision problem. Specifically, the i^{th} alternative — where $i = \operatorname{argmax}\left(\boldsymbol{p}_{\text{target}}\right)$ — is the most-preferred. Linearly combining priority vectors obtained from comparative judgements about the degree of influence by hierarchy-model elements with measurements against each element for each alternative produces the strengths of preference in $\boldsymbol{p}_{\text{target}}$.

The vector $\boldsymbol{p}_{\text{target}}$ does not contain expected utilities. The principles of decision science lead us to select the alternative producing the *greatest expected utility* (Kochenderfer (2015)). Chosing any other alternative constitues a departure from rationality. Abbas (2026) leveled this core criticism against decision methods based on "pairwise comparisons".

One might plausibly construct an AHP application so that $\boldsymbol{p}_{\text{target}}$ corresponds to a utility function, if not represnets an actual utility function itself. This must also employ a comparative-judgement intensity-of-importance assignment schema $\mathcal{J}$ (2a) consistent with the principles of probabilism (Pettigrew (2016)). Saaty (1986) did not provide for these by default. Since $\left[\boldsymbol{p}_{\text{target}}\right]_i \in [0,1]\ \forall i$, it coincidentally exhibits at least one property of a probablity.

Decision-Quality consequences of AHP. DQ emphasizes *Sound Reasoning* as a key link (Figure 1). Howard & Abbas (2016) and Spetzler, Winter, Meyer (2016) articulate decision-science foundations for decision quality in particular, and the decision-analysis discipline more generally. This is grounded in the von Neumann-Morgenstern principle that the most-rational decision selects the alternative yielding the greatest expected utility.

AHP in contrast returns a preferential ranking based on linear combination of comparative judgements and measurements of decision elements. Preferential ranking is not generally equivalent to identifying the maximum expected utility. Consequently no guarantee occurs that the most-preferred alternative also yields the expected maximum utility (Kochenderfer (2015)).

Identifying mechanics for an expected-utility-based AHP formulation lies beyond the scope of the present investigation. Reiterating the epistemic variance between AHP and mainstream decision science suffices here. Providing an information-theoretic characterization of the degree of influence by evidentiary elements within hierarchy models represents our goal. Consequently, the methodology developed here complements utility-optimizing decisional reasoning. Nothing herein should be construed as asserting that AHP sufficies as a sound reasoning approach in DQ terms.

Prioritized Titanium-alloy alternatives.

Table 4a contains the "raw" measurements of elements within hierarchy models $\left\{\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9}\right\}$ for each Titanium-alloy alternative considered by Zeng, D., et al (2025). The *source* column indicates location within Zeng, D., et al (2025) from which each element measurement was drawn. Exact values appear for each metallurgical element $\{B_1, \cdots, B_8\}$. No explicit measurements for hierarchy model B_9 appear. Figure 23 from Zeng, D., et al (2025) contains ordinal ranking of the alternatives according to B_9. The "$e_{\mathfrak{H}_n}$ effect" — *beneficial* versus *deleterious* — column in Table 4a is based on Table 2 from Zeng, D., et al (2025).

That the ***units*** column differs for each constituent hierarcy model $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9}$ drives home the entire point of MCDM. We seek a decision that weighs qualitatively different evidentiary elements. In rare cases — exemplified by Bu, et al (2026) — one might construct some computational model to combine the effects of qualitatively distinct evidence. Even when possible, the cost to obtain the resulting information might exceed its value.

Table 4b provides the normalized hierarchy-model element measurements. It results from applying the schema in (11b) to the rows in Table 4a, conditioned on the *effect* value associated with each metallurgical criterion.

For compactness, neither Table 4a nor Table 4b is represented in matrix form of (10a) and (11e), respectively. Obviously, one transposes each row from a table into a column vector in order to construct the block matrices. Subsequent calculations employ a matrix structure.

The rows of Table 4b contain elements of $\boldsymbol{p}_{\mathfrak{H}_i}$ corresponding to (11e). $\boldsymbol{P}_{\mathfrak{H}}$ then is the transpose of Table 4b. Obviously, Table 4b conforms to the requirements of (11c) and (11d). Moreover, $\boldsymbol{\Omega}_{\mathfrak{H}}$ is constructed from Table 2 as

$$\begin{aligned}\boldsymbol{\Omega}_{\mathfrak{H}} &= \begin{bmatrix} \boldsymbol{\omega}_{B_1} & \boldsymbol{\omega}_{B_2} & \cdots & \boldsymbol{\omega}_{B_9} \end{bmatrix} \\ &= \begin{bmatrix} 0.2582 & 0.1047 & \cdots & 0.2583 \\ 0.1047 & 0.2582 & \cdots & 0.1047 \\ 0.6370 & 0.6370 & \cdots & 0.6370 \end{bmatrix}. \end{aligned} \tag{13}$$

After populating the elements of $\boldsymbol{\omega}_{TA_{\text{opt}}}$ from Table 2, (12b) produces elements of $\boldsymbol{p}_{TA_{\text{opt}}}$ displayed in Table 5.

Table 5 represents the ultimate result: Preference-ranked Titanium-alloy alternatives

$$TA_{36} \succ TA_3 \succ TA_{10}. \tag{14}$$

A strong preference for TA_{36} appears within this alternative set. We prefer TA_{36} nearly four times as much

Table 4a: Metallurgical-property measurements $\boldsymbol{R}_{\mathfrak{H}}$ influencing Titanium-alloy prioritization. The *source* column indicates the exhibit within Zeng, D., et al (2025) that contains the element measurements for $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_9}$.

hierarchy model, $\mathfrak{H}$	element $e_{\mathfrak{H}_n}$	TA_3	TA_{10}	TA_{36}	units	$e_{\mathfrak{H}_n}$ effect	source
B_1	tensile strength	564	487	666	MPa	beneficial	Figure 10
B_2	impact energy	8.27	7.29	9.02	kN	beneficial	Figure 11
B_3	bending strength	848.9	742.0	1101.0	MPa	beneficial	Figure 13
B_4	surface hardness	200.5	200.4	239.57	HV	beneficial	Figure 14
B_5	friction-wear rate	0.06	0.17	0.03	$\%/h$	deleterious	Figure 15
B_6	erosion rate	0.36	0.37	0.25	mg/g	deleterious	Figure 17
B_7	corrosion current	3.90	4.39	5.04	$10^7 A/cm^2$	deleterious	Table 8
B_8	tribo-corrosion rate	100.2	177.5	59.63	μm	deleterious	Figure 22
B_9	material price	2	3	1	unitless	deleterious	Figure 23

Table 4b: Normalized metallurgical-property measurements $\boldsymbol{P}_{\mathfrak{H}}$ from applying (11b) to Table 4a.

	TA_3	TA_{10}	TA_{36}
B_1	0.4302	0	1
B_2	0.5665	0	1
B_3	0.2979	0	1
B_4	0.002	0	1
B_5	0.7857	0	1
B_6	0.0833	0	1
B_7	1	0.5702	0
B_8	0.6555	0	1
B_9	0.5	0	1

Table 5: Strengths of preference $\boldsymbol{p}_{TA_{\text{opt}}}$ from (12b) for candidate Titanium alloys considered by Zeng, D., et al (2025).

Titanium alloy alternative	strength of preference from (12b)
TA_{36}	0.5936
TA_3	0.1531
TA_{10}	0.0105

as we prefer TA_3. We in turn prefer TA_3 nearly fifteen times as much as we prefer TA_{10}.

By design, $\left[\boldsymbol{p}_{TA_{\text{opt}}}\right]_i \in [0,1]\ \forall i$. The strength of preference TA_{36} clarly falls in the upper half of this scale. To reiterate however, a link between the strength of preference and a business utility does not explicitly appear.

Additionally, Table 1 in Zeng, D., et al (2025) samples the Titanium-alloy chemical-composition sample space very sparsely. The alloy alternatives' composition consist of mixtures of nine distinct elements including Titanium. That TA_{36} produces a dramatically higher strength of preference by no means implies that it represents a global optimum.

Physical-model simulation along the lines of Bu, et al (2026) might not be practical for Zeng, D., et al (2025). However, the *Taguchi method* is often applied to efficiently localize optima in highly dimensional attribute spaces. Lin, N. et al (2019) exemplified this approach for a different material-design problem. Such an approach might lead to a higher-utility alternative in the neighborhod of alloy TA_{36}'s chemical composition.

Entropy of comparative-judgement graphs.

Our attention now pivots to information-theoretic analysis of the hierarchical framework comprised of sets of AHP comparative judgements. This constitutes our essential analytical innovation. Within AHP applications, strength-of-preference sensitivity analysis based on the linear operation (12b) represents common practice. That the graph structure resulting from combining comparative judgements using hierarchy models provides greater contrast provides our motivation.

We begin with brief consideration of the comparative-judgement graphs based on Tables 1a and 1b. Our attention then turns to Laplacian-matrix formulation for the different comparative-judgement scenarios. Within the context of the Titanium-alloy case study, Laplacians for $\mathfrak{H}_{B_1}, \cdots, \mathfrak{H}_{B_1}$ are straightforward, involving simple extensions of the approach from Brown, et al (2020). The graph associated with $\mathfrak{H}_{TA_{\mathrm{opt}}}$ however involves greater complexity, involving a mixed graph.

Following Brown, et al (2020) and others, Laplacian matrices provide the bases for graph-entropy calculations. Graph entropies for individual hierarchy-model comparative judgements follow straightforwardly. Coupling them into an overarching network involves innovative coupling using hierarchy-model priority vectors. That the latters' coincidence with the unit simplex proves useful, allowing us to treat the hierarchy-model tree graphs as probabilities.

We finally evaluate relative entropy to consider the loss of information — increase in entropy — occuring when a key informational component is omitted. Within the Titanium-alloy case study, the metallurgical criteria in Figure 2 represent the key informational components for which we characterize relative contributions. The approach by Jiroušek, Kratochvíl, Shenoy (2023) for belief-function graphical models informs our approach.

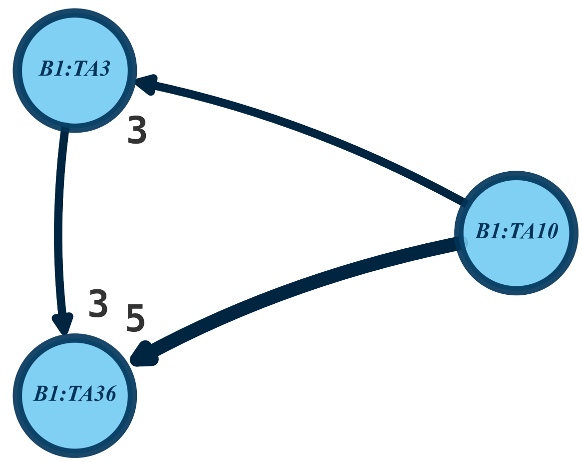
Construct comparative-judgement graphs.

Visualization of our comparative-judgement graphs provides useful preparation for construction of associated Laplacians, the prerequisite for entropy calculations. Here, we inspect visual representations of exemplary graphs. The depictions in Figures 4a and 4b are constructed using methods from the python *networkx* library (Hagberg, Schult, Swart (2008)), which invoke *matplotlib* methods (Hunter (2007)) to actually construct the figures.

The two figures conform to a common schema. Directed edges are represented by solid, dark-blue #002540 curves, and have arrow heads indicating their directivity. The thickness of the curve correlates with the *intensity of importance* assigned with the corresponding comparative judgement. Dashed, dark-red #23060b without arrow heads represent undirected edges.

Criterion-alternative comparative-judgement graphs. Figure 4a depicts $\mathcal{G}_{B_1}$, the directed-graph representation of hierarhcy model $\mathfrak{H}_{B_1}$. Table 1b lists the edges of which $\mathcal{G}_{B_1}$ is comprised. This corresponds to one of nine, criterion-alternative comparative judgements. Having only three nodes and three edges, these graphs are near-trivial. Explicitly depiction of graphs $\mathcal{G}_{B_2}, \cdots, \mathcal{G}_{B_9}$ provides little additional information. In Figure 4a, explicit depiction of the intensities of importance — the edge weight — is not inconvenient.

Figure 4a: Comparative-judgement graph $\mathcal{G}_{B_1}$ for tensile-strength metallurgical alloy-selection criterion based on hierarchy model $\mathfrak{H}_{B_1}$ for which edges are specified in Table 1b.



Note from Figure 4a and Table 1b that intensity-of-importance transitivity (6a) is not satisfied for $\mathfrak{H}_{B_1}$. We specifically find

$$a_{TA_{10},TA_3} = 3, \quad a_{TA_3,TA_{36}} = 3 \quad \text{and} \quad a_{TA_{10},TA_{36}} = 5 \; .$$

From Table 3, moreover, $CR_{B_1} \approx 0.039 < 0.1$, which satisfies the "good-enough" heuristic from Pant, et al (2022). This illustrates a relative insensitivity to deviations from (6a).

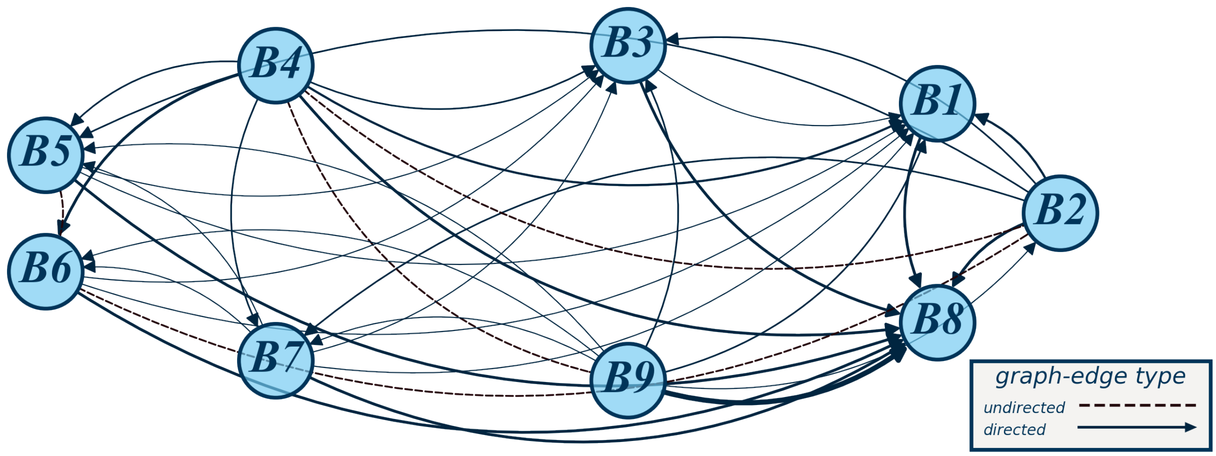
Target-criterion comparative-judgement graph. Figure 4b displays $\mathcal{G}_{TA_{\mathrm{opt}}}$, the graph for the target-to-criterion hierarchy model $\mathfrak{H}_{TA_{\mathrm{opt}}}$. The corresponding comparative-judgement edge list appears in Table 1a. The figure exemplifies the challenge of depicting higher-order graphs with high-valence vertices. A complete graph of order nine, $\mathcal{G}_{TA_{\mathrm{opt}}}$ makes for a messy plot.

As with Figure 4a, the edge weights correspond to the thickness of their corresponding curves in Figure 4b. An illegible figure results from explicitly depicting the edge weights here. The presence of four undirected edges in $\mathcal{G}_{TA_{\mathrm{opt}}}$ introduces a modicum of complexity into subsequent Laplacian-matrix construction.

Calculate entropies of comparative-judgement networks.

A variety of graph-entropy approaches exist. Sun, et al (2021) classified approaches based on graph-invariance characteristics of interest.

Figure 4b: Comparative-judgement graph $\mathcal{G}_{TA_{\text{opt}}}$ based on target-to-criterion hierarchy model $\mathfrak{H}_{TA_{\text{opt}}}$ for which edges are specified in Table 1a.



- *Vertex information-functional-based entropy* evaluates the robustness of the network and measures the importance of the vertex based on a vertex-centered information functional.

- *Distance-based entropy* evaluates classical topological indicators of interest in, e.g., mathematical chemistry.

- *Subgraph-based structure entropy* describes the information metrics of a subgraph to investigate the overall properties of the graph.

- Most-extensively explored, *eigenvalue entropy* provides indices charaterizing the structure, function, and evolution of a network.

We focus here primarily on importance of vertices. We seek to answer the question, "What elements of information — represented by graph vertices — most influence the utility of our alternatives?"

The literature contains a diverse variety of graph-entropy approaches. The majority involve calculating the spectra — Perron-eigenvalue arrays — associated with matrix representations of graphs' connectivity structures. The methods tend to be *ad hoc*: Few begin from some axiomatic foundation. Many are tailored to address the needs of a specific research context.

Bauer (2012) provides well-suited approach for purposes of this investigation. It generalizes the ubiquitous undirected-graph Laplacian (Godsil & Royle (2001); Biggs (2012); Brouwer & Haemers (2012)) to directed graphs with positive and negative edge weights. The approach straightforwardly extends to DAGs and mixed graphs.

Bauer's (2012) Laplacian formulation circumvents difficulties with a prominent approach by Chung (2005) based on transition-probablity matrices. Consistent with Perron-Frobenious theory, transition-probability spectra degenerate for DAGs, because terminal-state distributions concentrate into a single "sink" vertex. Alternatively, Chen, et al (2015) and Kuo (2022) propose entropies based solely on the graph weights and disregard the graph's connective structure. Bauer's (2012) Laplacian approach acounts for connectivity.

Data availability.

This GitHub repository ahp-graph-entropy contains the data and the python and Neo4j logic on which this paper is based.

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